

Michaelis-Mentin Approximation of Antibody Transport

To what extent does competition from previously generated, non-specific antibodies interfere with newly generated responses

```
#install.package("here")  
library(here)
```

here() starts at /Users/chirst/Desktop/Git/constraints-IgA

```
#install.package("ggplot2")  
library(ggplot2)  
  
#install.package("gridExtra")  
library(gridExtra)  
  
#install.packages("tidyverse")  
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
v dplyr      1.1.4      v readr      2.1.4  
v forcats    1.0.0      v stringr    1.5.1  
v lubridate  1.9.3      v tibble     3.2.1  
v purrr      1.0.2      v tidyr      1.3.0
```

```
-- Conflicts ----- tidyverse_conflicts() --  
x dplyr::combine() masks gridExtra::combine()  
x dplyr::filter()  masks stats::filter()  
x dplyr::lag()     masks stats::lag()  
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
#install.packages("deSolve")
library(deSolve)

#load functions
source(here("code", "functions", "functions.R"))

#choose plotting colors
my_colors = c("darkred", "maroon", "darkblue", "forestgreen", "darkorange", "goldenrod") #
num_colors = length(my_colors) #set number of colors needed for plotting
my_colors = colorRampPalette(my_colors)(num_colors) #increase number of colors staying in
```

Objective

In this document, we evaluate how long-lived IgA antibody titers remaining from previous infections of the respiratory mucosa interfere with the transport of newly generated, specific IgA antibodies to the site of infection. We assume that the concentration of M_{ns} changes negligibly over the course of infection, and assume that the steady state concentration of M_{ns} is proportional to $\frac{c}{d_\mu}$. We use a Michaelis-Menten approximation to calculate the flux of specific IgA across the mucosal membrane, and treat nonspecific antibodies as a competitive inhibitor of the transport process.

Outline

I will review the kinetics inferred by the Michaelis-Menten approximation before presenting the results of competitive inhibition by non-specific immunity during a specific response.

We are modelling the factors that contribute to the strength of competition between newly generated, pathogen specific IgA (M_s) and nonspecific IgA remaining from previous infections (M_{ns}) in the mucosa for transport across the epithelium and to the site of infection, the Lumen.

We are interested in how the half life of IgA titers in the mucosa - thus, the steady-state concentration of non-specific IgA in the mucosa - affects the magnitude and efficiency of newly-generated specific IgA responses. We therefore compare the effect of varying IgA half-life ($t_{1/2}$) on these two quantities to the effect of changing the rate of transport of antibodies across the epithelium (k_2) and effect of increasing the concentration of transport proteins (R_T) on the mucosal surface.

The results in this document are organized as follows:

- 1) How do $t_{1/2}$, k_2 , and R_T effect the flux of antibody into the lumen over the course of infection (VAR vs Time)
- 2) How do $t_{1/2}$ and k_2 affect i) the peak flux of the antibody and ii) the efficiency of the antibody transport to the lumen?
- 3) How do $t_{1/2}$ and R_T affect i) the peak flux of the antibody and ii) the efficiency of the antibody transport to the lumen?

Michaelis-Mentin Kinetics

Below, we define our parameters and vary the $t_{1/2}$ of mucosal antibody titers to observe how pre-existing antibodies increase the concentration of specific IgA at which point the transport of specific antibodies is half-saturated:

```
# transport
Vmax = 10^3 # order of magnitude of peak response # we might vary this later
k2 = 1/(2/24) #rate of transport - 1/T of transport which is 1/2hours or 1/2/24ths of a day
RT = Vmax/k2 #total number (concentration) of receptors, Vmax = k2RT
k1 = k2/10^3 #k2/min(Mns_vec) #rate of antibody binding such that KM is on the order of 10
KM = k2/k1 #checking KM

# steady state nonspecific antibodies
c = 1*10^5 #rate of influx of nonspecific IgA, Ab/time - comes from pb/dA - in mg/dl per day
t_halfs = c(14,30,90,400, 4000, 20000) #half lives of mucosal ab in days - 2 weeks, 1 month
d_mus = log(2)/t_halfs #half lives in days
names(d_mus) = t_halfs
Mns_vec = c/d_mus #vecotr of approximate ss solutions
names(Mns_vec) = t_halfs

# generation of new response
Ap = Mns_vec[round(length(t_halfs)/2)] #peak antibody titer will be the same as Mns for 90
```

Below, we plot how the flux of specific antibody for a given concentration of M_s in the mucosa changes as a function of the half-life of mucosal antibody titers. In essence, the effective amount of M_s required to reach half-maximal transport will increase by the concentration of non-specific antibody conferred by the increasing half-life (the effective $K_M = K_M + \hat{M}_{ns}$.)

```
#set concentrations of Ms for x axis
concentrations = 10^seq(1, 10, by = 0.1) #range of concentrations M_s such that we span mu
```

```

### plot flux vs [Ms] given each Mns
flux_df = lapply(t_halfs, function(t_half){
  # set d_mu and Mns
  d_mu = log(2)/t_half
  Mns = c/d_mu

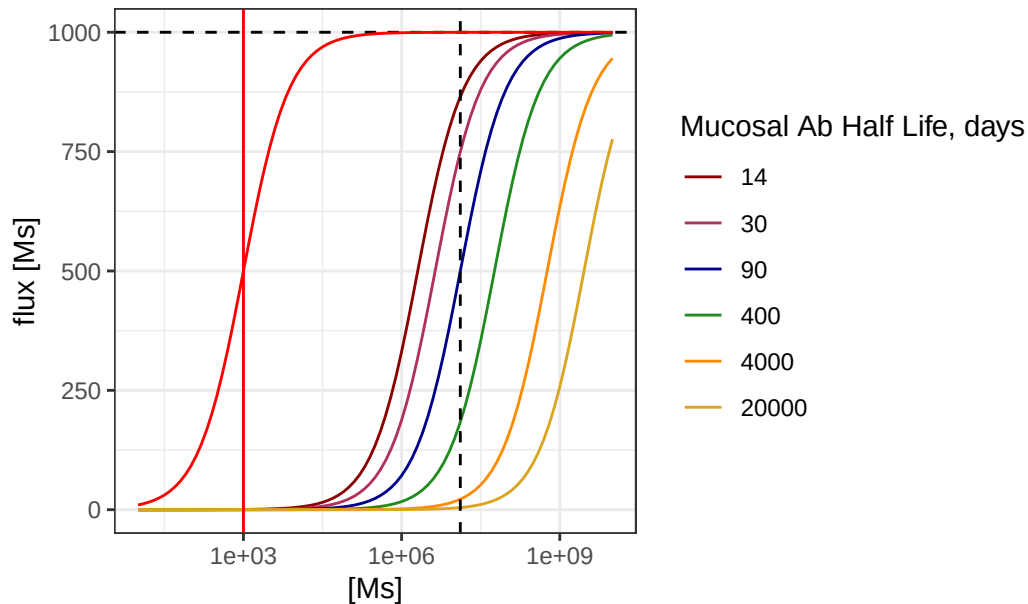
  #calculate flux using MM function in functions.R
  flux = sapply(concentrations, function(Ms){
    return(MM_function(Vmax = Vmax, KM = KM, M_s = Ms, M_ns = Mns))
  }) %>%
    cbind("Ms" = concentrations, "flux" = ., "t_half" = t_half) %>% # add the concentration
    as.data.frame() #format as a dataframe
}) %>%
  do.call(rbind, .) %>% as.data.frame() #combine all dataframes to make one long-form data

## plot
plot = ggplot() + geom_line(data = flux_df, aes(x = (Ms), y = flux, group = factor(t_half))
  geom_hline(yintercept = Vmax, linetype = "dashed") + #add vmax as horizontal line
  geom_vline(xintercept = KM, color = "red") + #add half-saturation point in absence of competition
  geom_vline(xintercept = Ap, linetype = "dashed") + #add vertical line showing peak titer
  geom_line(
    aes(x = concentrations, y = sapply(concentrations, function(Ms){MM_function(Vmax = Vmax,
    inherit.aes = FALSE,
    color = "red"
  ) + #add line showing how flux depends on Ms in the absence of competition
  scale_x_continuous(trans = "log10") + #x axis as log scale
  scale_color_manual(values = my_colors) + #set colors to my color scheme
  labs(x = "[Ms]", y = "flux [Ms]", color = "Mucosal Ab Half Life, days", title = "Competitive Inhibition")
  theme_bw() #changes background color and grid lines

# print to viewer
print(plot)

```

Competitive inhibition of specific antibody transport by nonspe



The horizontal dashed line shows the greatest possible transport rate, V_{max} ; the red curve shows the MM kinetics given $[M_s]$ only; the vertical red line shows the 50% saturation concentration given $[M_s]$ only, K_M ; and I have added a vertical dashed line to show the peak concentration of M_s generated over an infection.

Notice that I have chosen parameters such that, in the absence of competition, the maximum flux is reached by the the maximum concentration of M_s generated during a response. As nonspecific antibody is added, however, the flux achieved at the maximum specific antibody concentration drops. This is the core of the competitive effect.

Factors that affect the flux of antibody into the lumen

The competitive affect can be reduced by increasing the rate of transport of antibody across the membrane, as well as increasing the maximum transport rate, V_{max} . Both of these will increase the flux of specific antibody as well as nonspecific antibody across. Below, I plot how both of these factors change the flux of M_s in the presence of nonspecific antibody. I set the half life = 90 days for this example.

```
# transport
RT = 10^3/(1/(1/24)) #total number (concentration) of receptors, Vmax = k2RT
k2_vec = 1/(c(0.25,0.5,0.75,1,2,3)/24)
Vmax = k2_vec*RT #Vmax increases as well as KM when k2 increases
```

```

k1 = k1 # use k1 defined earlier
KM_vec = k2_vec/k1

# steady state nonspecific antibodies
c = 1*10^5 #rate of influx of nonspecific IgA, Ab/time - comes from pb/dA - in mg/dl per d
t_half = 90 #half life of mucosal ab in days
d_mu = log(2)/t_half #half life in days
Mns = c/d_mu #approximate ss solution

# generation of new response
Ap = Mns_vec[round(length(t_halves)/2)] #peak antibody titer will be the same as Mns for 90

#set concentrations of Ms for x axis
concentrations = 10^seq(1, 10, by = 0.1) #range of concentrations M_s such that we span mu

### plot flux vs [Ms] given each Mns
flux_df = lapply(k2_vec, function(k2){
  # set vmax and KM
  Vmax = k2*RT
  KM = k2/k1

  #calculate flux using MM function in functions.R
  flux = sapply(concentrations, function(Ms){
    return(MM_function(Vmax = Vmax, KM = KM, M_s = Ms, M_ns = Mns))
  }) %>%
    cbind("Ms" = concentrations, "flux" = ., "k2" = k2) %>% # add the concentrations of Ms
    as.data.frame() #format as a dataframe
}) %>%
  do.call(rbind, .) %>% as.data.frame() #combine all dataframes to make one long-from data

## plot
plot = ggplot() + geom_line(data = flux_df, aes(x = (Ms), y = flux, group = factor(k2), color = "red"),
  geom_vline(xintercept = (KM), color = "red") + #add half-saturation point in absence of
  geom_vline(xintercept = (Ap), linetype= "dashed") + #add vertical line showing peak tite
  geom_line(
    aes(x = concentrations, y = (sapply(concentrations, function(Ms){MM_function(Vmax = 10^3
    inherit.aes = FALSE,
    color = "red"
  ) + #add line showing how flux depends on Ms in the absence of competition
  scale_x_continuous(trans = "log10") + #x axis as log scale
  scale_color_manual(values = my_colors) + #set colors to my color scheme

```

```

labs(x = "[Ms]", y = "flux [Ms]", color = "k2, days-1", title = "Competitive inhibition
theme_bw() #changes background color and grid lines

# print to viewer
print(plot)

```

